

Information Retrieval: Assignment 3

Retrieval Evaluation

Introduction

In this assignment, you will delve into the implementation of fundamental retrieval evaluation metrics. These metrics will allow you to critically evaluate and compare the results obtained from different search engines, and help you to strengthen a deeper insight into the intricacies of information retrieval techniques.

You are provided with a `main.py` file with the TODOs and cached search results in JSON format of four different search engines:

- Bing
- DuckDuckGo
- Google
- Yahoo

There are two directories for two different search queries:

query1__cache: Modern Information Retrieval

query2__cache: information retrieval evaluation

How to run the provided code

The code was tested using Python version 3.10 and you can run it using multiple tools:

Docker

You are provided with a `Dockerfile` and a `docker-compose.yml`. The command to run:

```
docker compose up --build
```

`-build` ensures that if any changes happened the images gets rebuilt, you should not need this all the time but always include it to make sure.

Without Docker

You can also run the code without Docker.

virtualenv

First make sure you have `virtualenv` installed then run:

```
virtualenv venv
```

```
source venv/bin/activate
```

```
pip install -r requirements.txt
```

```
python main.py
```

without virtualenv

```
pip install -r requirements.txt
```

```
python main.py
```

Important Note

It is probably a good idea to redirect your output to a `results.txt` file, you can do so by

```
python main.py > results.txt
```

which is also what the Dockerfile does. Additionally, the graphs will be saved to a directory called `plots`.

Finally, the reference plots will be available on Themis, you can find them here: themis.housing.rug.nl/course/2025-2026/inforet/ass3.

Additionally, we will be putting some results on Themis, so you can submit your code to test whether it works. Say Themis has an issue, you can always and is encouraged to look at the expected out and compare that to your output. Remember that passing on Themis is not required.

Description

Implement fundamental retrieval evaluation metrics. Use these metrics to evaluate the retrieved results of the 4 search algorithms (this also includes the baseline/reference algorithm).

Precision and Recall

- Implement a method to compute the precision and recall.
- Evaluate the ranking results produced by the 4 search algorithms.

Precision vs Recall

- Plot the precision figure for the 11 standard recall values.
- Evaluate the ranking results produced by the 4 search algorithms.
- Pay attention to the boundary cases (refer to the example queries q1 and q2 from the lecture).

Single valued summaries

- Implement the metric Precision at rank 5 ($P@5$), Precision at rank 10 ($P@10$), and the F-measure.
- Evaluate the ranking results produced by the 4 search algorithms.

Deliverables

- Code + Documentation
- A brief report comparing the evaluation of the three search algorithms. For each metric briefly describe the metric, the scores/plots obtained on the three search techniques, and a comparative evaluation of these scores.